# Financial Mathematics
# (MATH35400/MATHM5400)

**Problem Class 5 – 20 February 2026** [1]

**Problems:**

1. (2020 exam, level 3, Q2c) Consider a model with two periods and one risky security where the interest rate is $r = 0$ and the price process is given by the following table:

| State | $t=0$ | $t=1$ | $t=2$ |
|---|---|---|---|
| $\omega_1$ | 10 | 16 | 18 |
| $\omega_2$ | 10 | 16 | 12 |
| $\omega_3$ | 10 | 8 | 12 |
| $\omega_4$ | 10 | 8 | 6 |

We consider two options:

- An *Asian* option with strike price 14 expiring at time 2, so that its payoff is given by
$$X = \left(\frac{S_0 + S_1 + \cdots + S_T}{3} - 14\right)^+.$$
- A *look-back* option with strike price 14 expiring at time 2, so that its payoff is given by
$$Y = \left(\max_{i=0,\dots,T} S_i - 14\right)^+.$$

(a) Calculate the risk-neutral probability measure $\mathbb{Q}$ and specify $\mathbb{Q}(\{\omega_i\})$ for $i = 1, \dots, 4$.

(b) Determine $X(\omega_i)$ and $Y(\omega_i)$ for $i = 1, \dots, 4$.

(c) Determine the risk-neutral prices of $X$ and $Y$ at time $t = 0$.

(d) Find replicating portfolios for each of the claims $X$ and $Y$.

2. (2018 exam, Q2 (b ii)) Consider a two period model with zero interest rate and a single risky asset whose price is given in the table below.

| $\omega_k$ | $t=0$ | $t=1$ | $t=2$ |
|---|---|---|---|
| $\omega_1$ | $S_0 = 6$ | $S_1 = 9$ | $S_2 = 10$ |
| $\omega_2$ | $S_0 = 6$ | $S_1 = 9$ | $S_2 = 7$ |
| $\omega_3$ | $S_0 = 6$ | $S_1 = 3$ | $S_2 = 7$ |
| $\omega_4$ | $S_0 = 6$ | $S_1 = 3$ | $S_2 = 1$ |

This model is complete and $\mathbb{Q}(\{\omega_1\}) = 1/3$, $\mathbb{Q}(\{\omega_2\}) = 1/6$, $\mathbb{Q}(\{\omega_3\}) = 1/6$, $\mathbb{Q}(\{\omega_4\}) = 1/3$ is a risk-neutral measure.

A *double* European option gives the holder the right to *buy or sell* the risky asset at time $t = 2$ for price $K$. Write down the pay-off function for this option in terms of $K$, then calculate its time zero price by risk-neutral valuation in the case $K = 7$.